Medical Microbiology

Lecture Thirteen

Gram Negative diplococcic

Third stage- College of Dentistry

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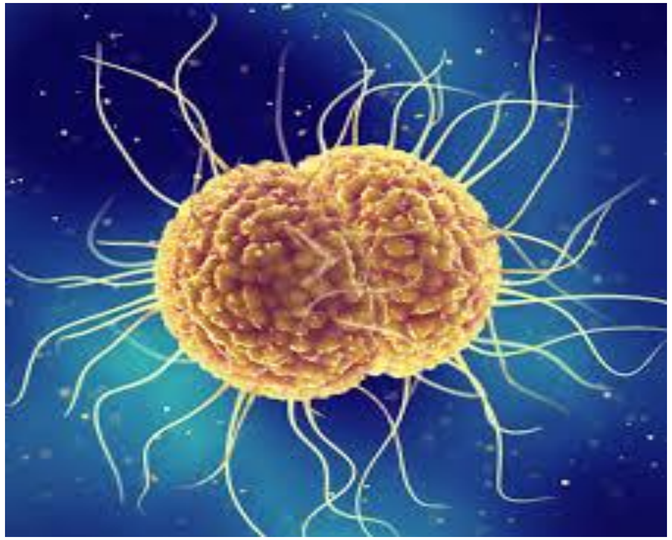
Introduction:

- Gram-negative diplococcic are a significant group of bacteria that typically appear in pairs (diplococcic) under the microscope.
- Diplococcic mean Spherical bacteria that are arranged in pairs.
- These microorganisms play a crucial role in medicine and microbiology.
- *Neisseria* is example of this group.



General Characteristics of Neisseria :

- Bacteria it appear as (G-)gram negative, arranged in pairs, coffee bean shaped diplococcic, non motile, non sporing forming.
- Microbial Properties: Do not form chains or clusters.
- Colonies on blood and chocolate agar appear mucosal surface , gray, smooth, rounded, and non hemolytic.
- Incubation at (35-37)C for 48 hours and in an atmosphere containing (5-10)% Co2.

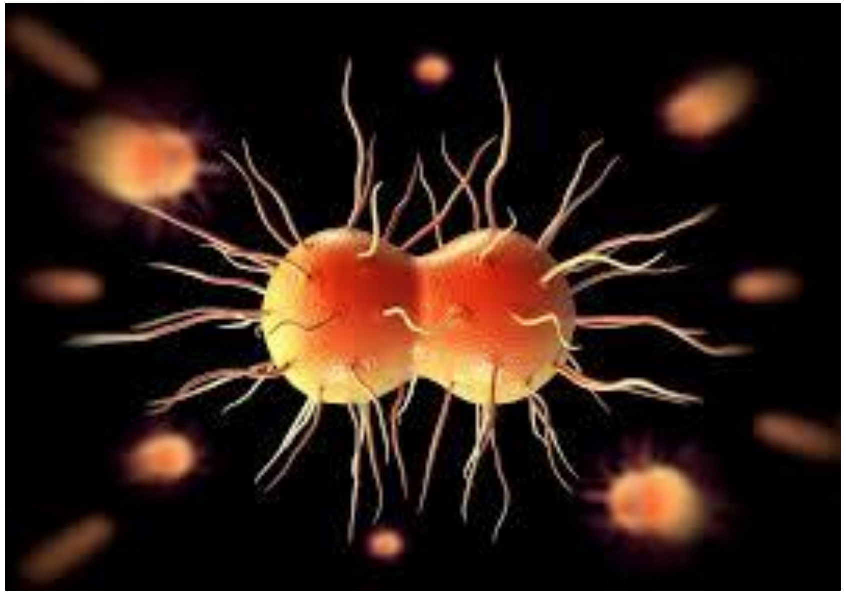
Common Examples

1. Neisseria meningitidis :

- Causes meningitis and nasopharynx are spread by contaminated respiratory droplets.

2. Neisseria gonorrhoeae :

- Causes gonorrhea (infection of urethra in men and cervix in women) transmitted through sexual contact.



Importance of Neisseria in Dentistry:

The genus **Neisseria** includes both pathogenic and commensal species. In dentistry, its importance is mainly related to the **oral micro-biota, biofilm formation,** and **opportunistic infections.**

- **Types of *Neisseria* in the Oral Cavity (Commensal Species):**

Common species found in the oral cavity and oropharynx include:

- ***Neisseria sicca***
- ***Neisseria mucosa***
- ***Neisseria subflava***
- ***Neisseria flavescens***

They are commonly present in **Saliva**, **Tongue surface**, **Early dental plaque** and **Oropharynx**.

Role in the Oral Ecosystem:

1-Early Colonization:

They are among the first bacteria to attach to the acquired pellicle on the tooth surface.

2-Biofilm Formation:

They contribute to the development of dental biofilm and facilitate the attachment of other microorganisms.

3-Ecological Balance:

They may help maintain microbial balance by competing with pathogenic bacteria.

Virulence Factors

1-Fimbriae (Common Pili): Hair-like structures that enhance adherence to host cells, facilitating colonization.

2-Lipooligosaccharide: A component of the outer membrane that contributes to immune evasion and inflammation.

3-Capsule: A protective layer that prevents phagocytosis, helping the bacteria evade the immune response.

4-Cell Membrane Proteins: Proteins that play roles in adhesion, invasion, and immune system evasion.

5-IgA Protease: An enzyme that cleaves IgA antibodies, allowing bacteria to resist mucosal immunity.

Symptoms of Meningitis:

1-General poor feeling.

2-Sudden high fever.

3-Sever headache.

4-Nausea or some time vomiting.

5-Discomfort in bright lights.

6-Joint pain.

Symptoms of gonorrhea:-

1-Discharge from the vagina(watery , creamy , slightly green).

2-Pain or burning sensation while urinating.

3-Urinate more frequently.

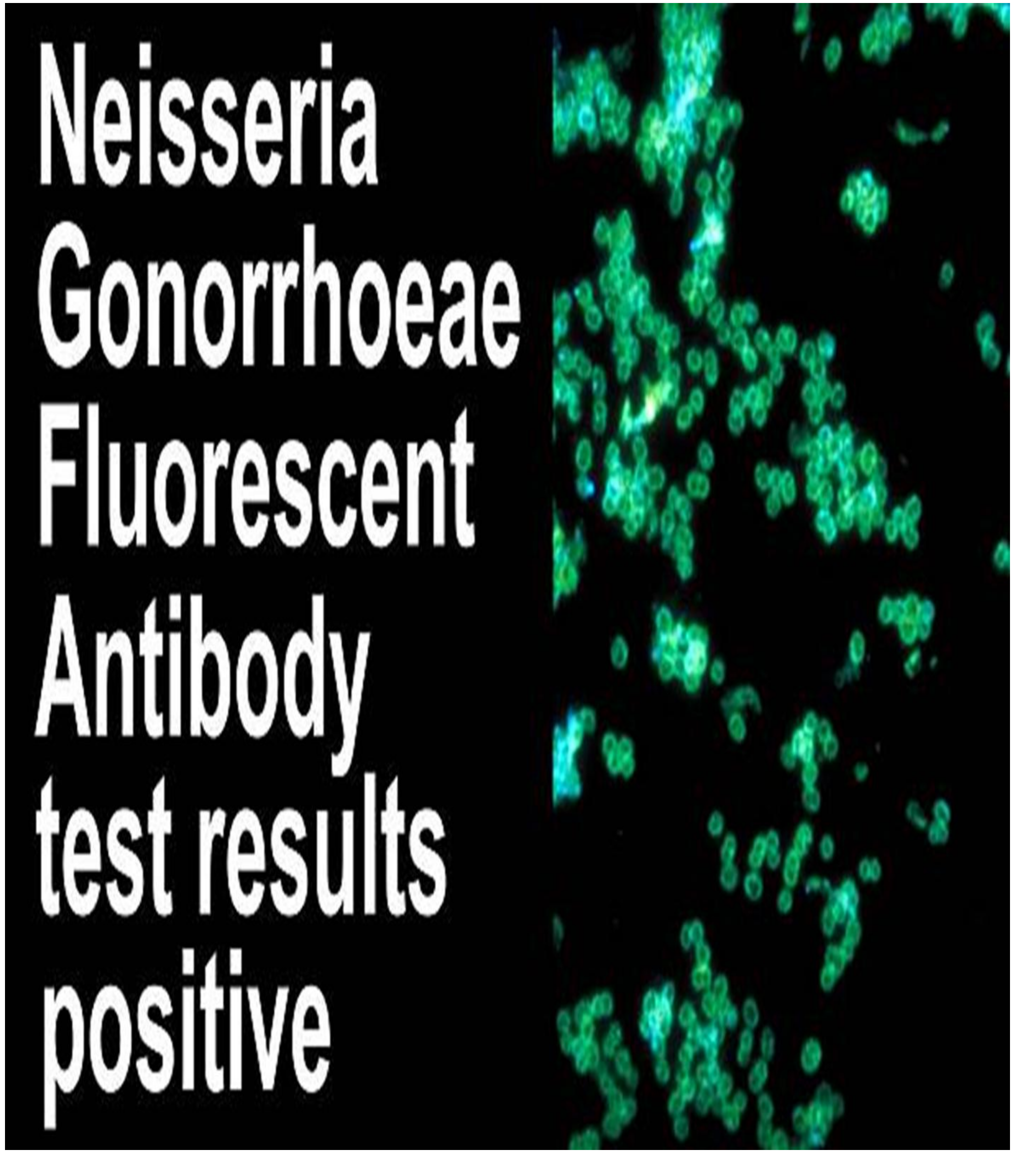
4-Sore throat.

5-Sharp pain in the lower abdomen.

6-Fever.

Methods of examination:-

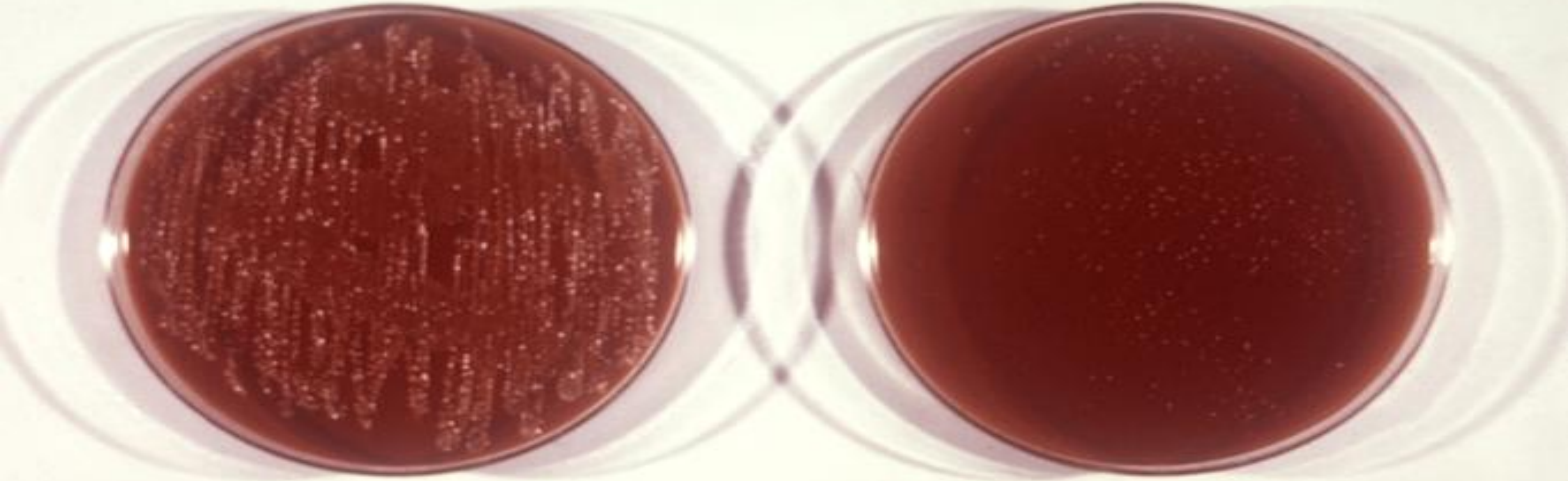
1-Direct microscopy (gram staining).
2-Culture.
3-Serological tests.
4-**Immunofluorescence**.
5-Biochemical tests.



Laboratory diagnosis:-

1-Gram Stain: Gram-negative, coffee bean-shaped diplococcic.
2-Growth on Thayer-Martin medium **(selective media)** chocolate agar with antimicrobials **Colistin** (to inhibit G- bacilli), **Vancomycin** (to inhibit G+ bacteria), and **Nystatin** (to inhibit Yeast).
3-Oxidase +ve.
4-Catalase +ve.
5-N.meningitidis Ferments **glucose** and **maltose** with acid production, while **N. gonorrhoeae** ferment only **glucose.**

Rectal Specimen
(Testing for *Neisseria gonorrhoeae*)



Chocolate Medium Overgrowth

Thayer-Martin Medium *Neisseria* Only

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Treatment of Neisseria:-

1-Ceftriaxone (1-2) g IV every (12-24) hours.
2-Penicillin.
3-Azithromycin.

Feature	N. meningitidis	N. gonorrhoeae
Capsule	Present (important virulence factor)	Absent
Microscopic Shape	Lens-shaped diplococci (flattened adjacent sides)	Kidney-shaped diplococci (concave adjacent sides)
Gram Stain	Gram-negative diplococci	Gram-negative diplococci
Sugar Fermentation	Ferments glucose & maltose	Ferments glucose only
Location in Smear	Intra- & extracellular	Mainly intracellular (inside neutrophils)
Colony Morphology	Circular, smooth colonies	Irregular margins
Habitat	Nasopharynx	Genital tract
Main Diseases	Meningitis, meningococcemia	Urethritis, cervicitis, Pelvic Inflammatory Disease

Any Question?

Thank you